Exp No:13 Experiment to handle missing values and inappropriate data in a given data set .

DESCRIPTION:

Understand the procedure to handle missing and inappropriate data

PROGRAM:

```
import numpy as np
import pandas as pd
# Load dataset
df = pd.read_csv("Restaurant_Dataset.csv")
print("Original Dataset:")
print(df)
# Remove duplicate rows
df.drop_duplicates(inplace=True)
# Handle inappropriate values
df.loc[df["CustomerID"] < 0, "CustomerID"] = np.nan
df.loc[df["Bill"] < 0, "Bill"] = np.nan
df.loc[df["EstimatedSalary"] < 0, "EstimatedSalary"] = np.nan
df.loc[(df["NoOfPax"] < 1) | (df["NoOfPax"] > 20), "NoOfPax"] = np.nan
df.loc[
   (df["Rating(1-5)"] < 1) | (df["Rating(1-5)"] > 5),
   "Rating(1-5)"
] = np.nan
# Correct inconsistent food preferences
df["FoodPreference"] = df["FoodPreference"].replace(
```

```
    ["vegetarian", "Vegetarian", "veg"], "Veg"
)
df["FoodPreference"] = df["FoodPreference"].replace(
    ["non-veg"], "Non-Veg"
)
# Display missing values before handling
print("\nMissing Values:")
print(df.isnull().sum())
# Fill missing values
df["EstimatedSalary"] = df["EstimatedSalary"].fillna(
    round(df["EstimatedSalary"].mean())
)
df["Bill"] = df["Bill"].fillna(
    round(df["Bill"].mean())
)
df["NoOfPax"] = df["NoOfPax"].fillna(
    round(df["NoOfPax"].median())
)
df["Rating(1-5)"] = df["Rating(1-5)"].fillna(
    round(df["Rating(1-5)"].median())
)
print("\nCleaned Dataset:")
print(df)
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())
```

OUTPUT:

```
Original Dataset:
    CustomerID    Bill  EstimatedSalary  NoOfPax  Rating(1-5) FoodPreference
0         1001   850.0          42000.0      2.0          5.0            Veg
1         1002  1250.0          55000.0      4.0          4.0        Non-Veg
2         1003   640.0          38000.0      3.0          4.0     vegetarian
3         1004  2100.0          72000.0      6.0          5.0        non-veg
4         1005   450.0          31000.0      1.0          3.0            veg
5         1006   980.0          46000.0      2.0          4.0     Vegetarian
6         1007  1600.0          68000.0      5.0          6.0        Non-Veg
7         1008  -250.0          51000.0      3.0          4.0            Veg
8        -1009   730.0          40000.0      2.0          3.0        Non-Veg
9         1010  1150.0          -3000.0      4.0          5.0            Veg
10        1011   890.0          44000.0      0.0          4.0        Non-Veg
11        1012  1350.0          59000.0     25.0          2.0            Veg
12        1013     NaN          48000.0      3.0          4.0        Non-Veg
13        1014   720.0              NaN      2.0          5.0     vegetarian
14        1015  1900.0          63000.0      NaN          4.0        non-veg
15        1016   560.0          35000.0      2.0          NaN            Veg
16        1017  1020.0          47000.0      3.0          4.0        Non-Veg
17        1018  1450.0          61000.0      5.0          5.0     Vegetarian
18        1019   830.0          39000.0      2.0          3.0        non-veg
19        1020  1750.0          70000.0      4.0          4.0            Veg
20        1002  1250.0          55000.0      4.0          4.0        Non-Veg

Missing Values:
CustomerID         1
Bill               2
EstimatedSalary    2
NoOfPax            3
Rating(1-5)        2
FoodPreference     0
dtype: int64

Cleaned Dataset:
    CustomerID    Bill  EstimatedSalary  NoOfPax  Rating(1-5) FoodPreference
0       1001.0   850.0          42000.0      2.0          5.0            Veg
1       1002.0  1250.0          55000.0      4.0          4.0        Non-Veg
2       1003.0   640.0          38000.0      3.0          4.0            Veg
3       1004.0  2100.0          72000.0      6.0          5.0        Non-Veg
4       1005.0   450.0          31000.0      1.0          3.0            Veg
5       1006.0   980.0          46000.0      2.0          4.0            Veg
6       1007.0  1600.0          68000.0      5.0          4.0        Non-Veg
7       1008.0  1123.0          51000.0      3.0          4.0            Veg
8          NaN   730.0          40000.0      2.0          3.0        Non-Veg
9       1010.0  1150.0          50500.0      4.0          5.0            Veg
10      1011.0   890.0          44000.0      3.0          4.0        Non-Veg
11      1012.0  1350.0          59000.0      3.0          2.0            Veg
12      1013.0  1123.0          48000.0      3.0          4.0        Non-Veg
13      1014.0   720.0          50500.0      2.0          5.0            Veg
14      1015.0  1900.0          63000.0      3.0          4.0        Non-Veg
15      1016.0   560.0          35000.0      2.0          4.0            Veg
16      1017.0  1020.0          47000.0      3.0          4.0        Non-Veg
17      1018.0  1450.0          61000.0      5.0          5.0            Veg
18      1019.0   830.0          39000.0      2.0          3.0        Non-Veg
19      1020.0  1750.0          70000.0      4.0          4.0            Veg

Missing Values After Cleaning:
CustomerID         1
Bill               0
EstimatedSalary    0
NoOfPax            0
Rating(1-5)        0
FoodPreference     0
dtype: int64
```